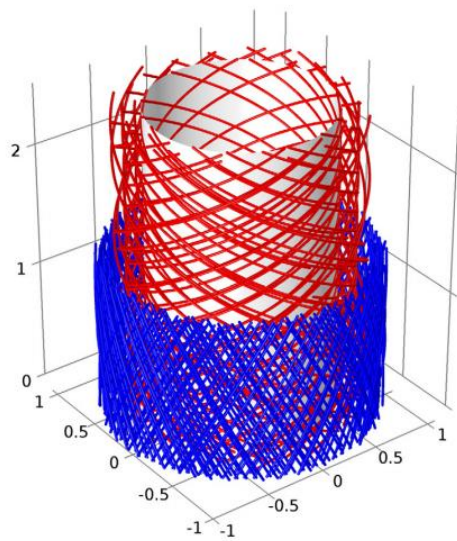


Arterial Wall Mechanics

A review of Constitutive Models and Implementation in COMSOL Multiphysics



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BME 552

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Table of Contents

1.0 Introduction	3
2.0 Structural Properties of Arterial Walls.....	3
2.1 Intima.....	4
2.2 Media	5
2.3 Adventitia	5
3.0 Mechanical Behavior of Arterial Tissue	6
3.1 Components of Mechanical Behavior	6
3.2 Typical Mechanical Behavior	6
4.0 Finite Hyperelasticity.....	7
4.1 Hyperelasticity Stress Response	7
4.2 Basic Kinematics	9
4.3 Equilibrium Equations	11
5.0 Constitutive Model for Arterial Walls	13
6.0 Results & Discussion.....	16

1.0 Introduction

The increasing interest in modeling arterial wall mechanics stemmed from Humphrey's comparative study on constitutive models to find the mechanical response of arteries. An explanation for this increase is the realization of mechanical forces influencing diseases such as atherosclerosis which has led to high mortality rates in the western world. To understand these mechanical forces, we require accurate constitutive models which describe how arterial tissue deforms under different loads.

Arteries have very complex structures and experience internal blood pressure and stresses that come from natural growth and remodeling of themselves. The authors show that normal models don't consider these features which affect their predictive abilities and present their new model which introduces collagen fibers to the structure of arteries.

The motivation behind a new and reliable model is to understand diseases such as atherosclerosis and capture the complex behavior of arteries under deformation and further improve simulations for clinical procedures.

2.0 Structural Properties of Arterial Walls

Understanding these challenging models requires an understanding of the background and structure of arterial walls. Arteries are composed of 3 layers: Intima, Media, and Adventitia. As shown in Figure 1, these 3 layers have their own composition and play their own mechanical functions in the models.

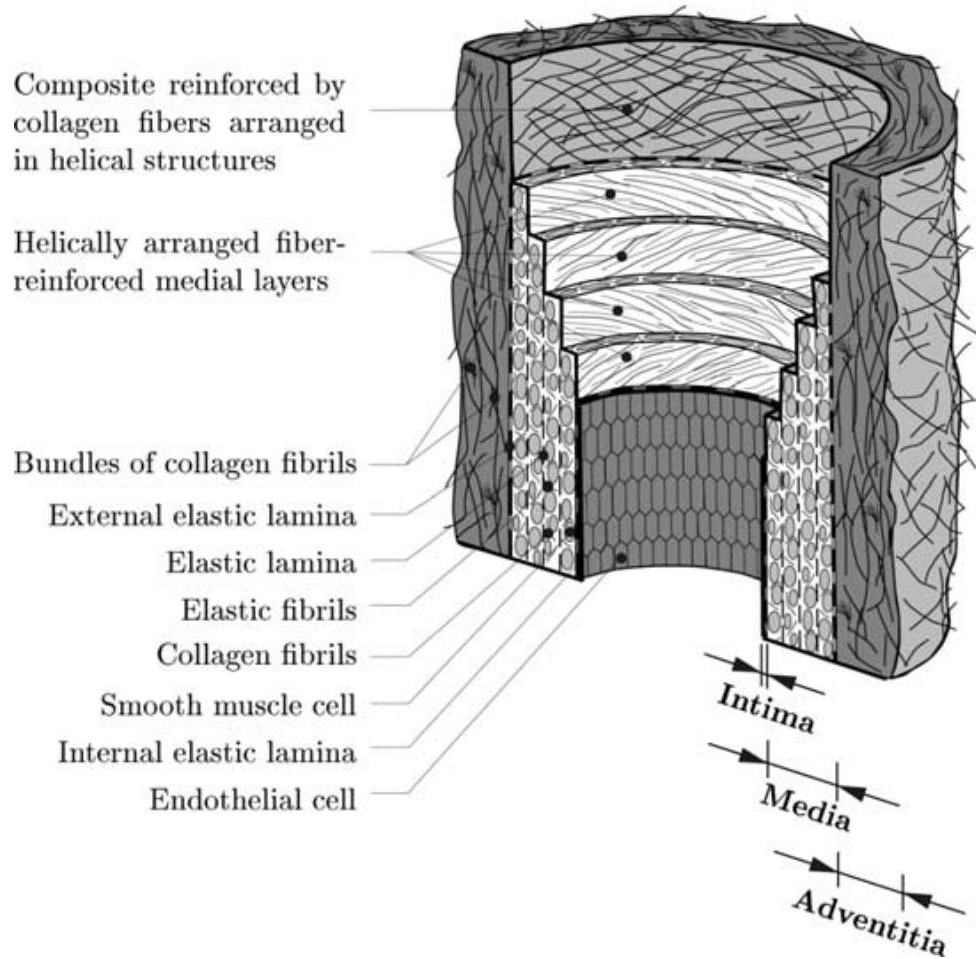


Figure 1: Model of the major components of a healthy elastic artery composed of 3 layers

2.1 Intima

Intima is the thin inner layer of the artery. Resting on a thin basal membrane, the intima is mostly made up of endothelial cells. Its role becomes important in conditions such as atherosclerosis for older, less healthy individuals where plaques accumulate in their arteries leading to a buildup and creating a stiff layer. For healthy and young individuals, it has minimal mechanical contributions.

2.2 Media

The media is the middle layer of the artery, and it plays an important role in the mechanical function. It is made up of collagen, elastin, and smooth muscle cells which are formed into a structurally complex layer. The elastin fibers help protect and restore the artery to its original shape after experiencing deformations. The collagen fibers on the other hand are what further stiffen the arteries after large strains. The media has this helical shaped fiber structure composed of reinforced fiber layers, which gives the artery its strength to handle different kinds of stresses.

2.3 Adventitia

Adventitia is the last outer layer of the artery, made up of primarily collagen fibers in a helical structure and loose connective tissue. It plays an important role in high pressure because it provides stiffness to the arteries, even though it is less stiff than the media at low pressures. The collagen fibers are influenced by the pressure on the arteries. For example, at high pressures the fibers straighten out and make the artery stiffer which helps stop it from tearing. Therefore, the mechanical aspect of the adventitia becomes important when the artery is under deformation. Together these 3 layers form a nonlinear and anisotropic composite structure that can experience different kinds of forces.

3.0 Mechanical Behavior of Arterial Tissue

3.1 Components of Mechanical Behavior

Elastin: dominates at low strains; provides elasticity.

Collagen: recruited at higher strains; provides stiffness, especially in the adventitia.

Ground substance and smooth muscle cells: contribute to viscoelastic and active behavior.

3.2 Typical Mechanical Behavior

Arteries show nonlinear stress-strain curves with exponential stiffening under high pressures. Figure 2 shows this stress-strain response with a circumferential arterial strip, located in the media, under uniaxial tension.

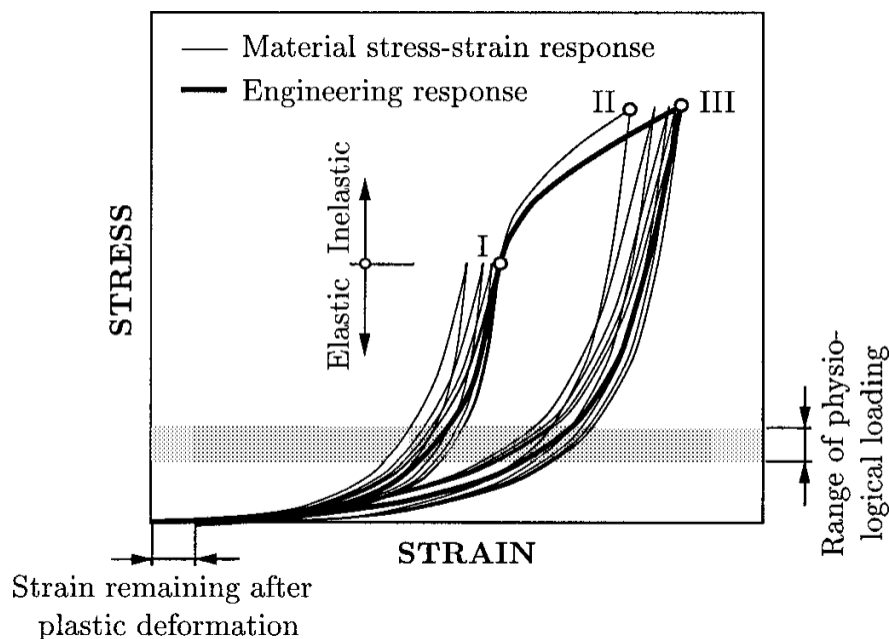


Figure 2: Stress-Strain response

Initially we notice a low stiffness region, during the cyclical loading and unloading where we see less resistance. Then as more collagen fibers are introduced, we see an elastic and exponential

increase in the curve as the tissue reaches point 1. When the tissue gets stretched to point 2 it experiences this inelastic deformation and won't return to its original state. It then stabilizes at point 3 which behaves like point 1, and the black line shows the general mechanical response of the tissue.

4.0 Finite Hyperelasticity

This part of the paper shows the mathematical principles used to describe the behavior of the artery subject to different kinds of loadings. Arteries are expected to behave like hyperelastic materials after large deformations and a way to show this can be derived from a strain energy function. The determinant $J = \det(\mathbf{F})$ is important for measuring the volume change during the large deformations. Then to modify the tensor to isolate the volume change from the shape change, the deformation gradient $\mathbf{F}$, is decomposed. This helps us use the modified Cauchy Green tensor $\mathbf{C}$ in eq (2), making it independent to volume changes.

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = J^{2/3} \bar{\mathbf{C}}, \quad \bar{\mathbf{C}} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}, \quad (2)$$

4.1 Hyperelasticity Stress Response

To describe stress in hyperelastic materials, the Helmholtz free energy function introduces important elements such as strain and structural tensors.

$$\Psi(\mathbf{E}, \mathbf{A}_1, \dots, \mathbf{A}_n) = U(J) + \bar{\Psi}(\bar{\mathbf{E}}, \mathbf{A}_1, \dots, \mathbf{A}_n), \quad (5)$$

In equation 5 we see U which represents volumetric response and the function Ψ represents the isochoric part.

The 2nd Piola- Kirchhoff stress tensor is introduced in eq (6):

$$\mathbf{S} = \mathbf{S}_{\text{vol}} + \bar{\mathbf{S}} \quad \text{with } \mathbf{S}_{\text{vol}} = \frac{\partial U(J)}{\partial \mathbf{E}}, \quad \bar{\mathbf{S}} = \frac{\partial \bar{\Psi}(\bar{\mathbf{E}}, \mathbf{A}_1, \dots, \mathbf{A}_n)}{\partial \mathbf{E}}. \quad (6)$$

The equation shows us how internal stress connects to the energy that is stored during deformation. We see repeat variables from previous equations as we are introduced to new variables such as $\mathbf{S}$ and $\mathbf{E}$. $\mathbf{S}$ represents the Second Piola–Kirchhoff stress tensor while the $\mathbf{E}$ is the Green–Lagrange strain tensor. The tensor becomes defined with respect to the reference configuration.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{vol}} + \bar{\boldsymbol{\sigma}} \quad \text{with } \boldsymbol{\sigma}_{\text{vol}} = p\mathbf{I}, \quad \bar{\boldsymbol{\sigma}} = J^{-1} \text{dev} \left(\bar{\mathbf{F}} \frac{\partial \Psi}{\partial \bar{\mathbf{E}}} \bar{\mathbf{F}}^T \right), \quad (10)$$

Then we are introduced to the alternative form of the equation for the Cauchy stress in the deformed configuration. Eq 10 is the result from the deformation gradient being used while transforming the Piola-Kirchoff stress (restated from eq, 8). The two parts of the equation involve hydrostatic pressure, a Lagrange multiplier when introduced as variable p , and then the fiber structure as it accounts for stretching and torsion. The operator $\text{dev}(\cdot)$ is the deviatoric operator and it removes the isotropic part of a tensor to focus on distortional stress, which is useful for shape changing tissue.

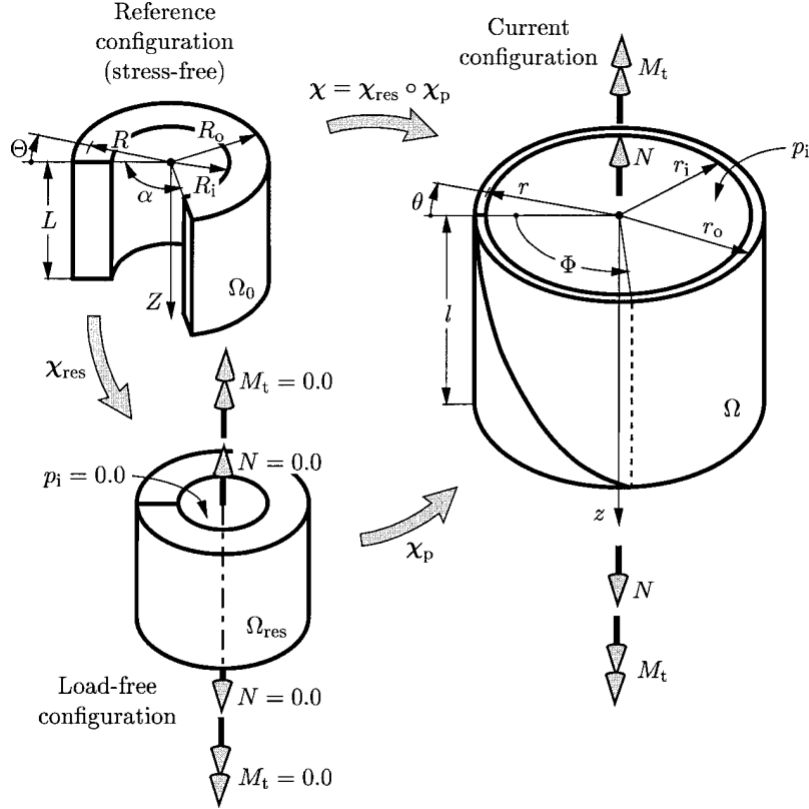


Figure 3. Arterial ring in the the (stress-free) reference configuration Ω_0 , the load-free configuration Ω_{res} and the current configuration Ω .

This figure shows how the arterial model starts in the stress-free and opened configuration and then when placed under bending it goes into the closed shape in the load free configuration after introducing residual stress. Further deformation such as stretching moves the artery into the final state. From this, we can see how stress changes through each step, from internal structure to residual stress and then to external loads which are shown by the equations above.

4.2 Basic Kinematics

The basic kinematics equations help us understand how arteries are influenced by mechanical loads. To figure this out, we look at how arteries deform in geometry or in motion as modeled in the article as a cylindrical tube. In its stress-free or reference configuration, the

deformations are measured with the help of the following reference geometry shown in equation 12:

$$R_i \leq R \leq R_o, \quad 0 \leq \Theta \leq (2\pi - \alpha), \quad 0 \leq Z \leq L, \quad (12)$$

Here R_i , R_o , α and L represent the inner and outer radius. This configuration helps us understand how the artery would behave in its open state or if it was cut open. To show how the artery would behave under loads in its functional state, the article maps the reference to deformed configuration that accounts for loads such as bending or torsion which is shown in equation 15.

$$r = \sqrt{\frac{R^2 - R_i^2}{k\lambda_z} + r_i^2}, \quad \theta = k\Theta + Z\frac{\Phi}{L}, \quad z = \lambda_z Z, \quad (15)$$

In this equation, r , θ , and z are the deformed coordinates, while k represents the arterial wall going from an open to closed shape. λ is the axial stretch ratio, and Φ represents the angle of twist applied over the length L in equations 15(2,3). Then from the mapping of its functioning state, the arterial stretching in each direction needs to be calculated, shown in the stretch equations below:

$$\begin{aligned} \lambda_r(R) &= \frac{\partial r}{\partial R} = \frac{R}{rk\lambda_z}, & \lambda_\theta(R) &= \frac{r}{R} \frac{\partial \theta}{\partial \Theta} = \frac{kr}{R}, \\ \gamma(R) &= r \frac{\partial \theta}{\partial z} = r \frac{\Phi}{L}. \end{aligned} \quad (16)$$

In equation 16 we see the radial, circumference stretching, and shear strain from torsion. These values are used to define the deformation gradient F , which shows the combination of stretching and shear in the arteries deformation.

$$\mathbf{F} = \bar{\mathbf{F}} = (\lambda_\theta \lambda_z)^{-1} \mathbf{e}_r \otimes \mathbf{E}_R + \lambda_\theta \mathbf{e}_\theta \otimes \mathbf{E}_\Theta + \gamma \lambda_z \mathbf{e}_\theta \otimes \mathbf{E}_Z + \lambda_z \mathbf{e}_z \otimes \mathbf{E}_Z, \quad (19)$$

The tensor shown in equation 19 repeats the previously stated stretches in the various directions through the e components. It is useful for calculating strain and stress tensors. Finally in

equations 20 and 21, we can understand where the right and left Cauchy-green deformation tensors were constructed from.

$$\begin{aligned} \mathbf{C} = \overline{\mathbf{C}} = & (\lambda_\theta \lambda_z)^{-2} \mathbf{E}_R \otimes \mathbf{E}_R + \lambda_\theta^2 \mathbf{E}_\Theta \otimes \mathbf{E}_\Theta + \lambda_z^2 (1 + \gamma^2) \mathbf{E}_Z \otimes \mathbf{E}_Z \\ & + \gamma \lambda_\theta \lambda_z (\mathbf{E}_\Theta \otimes \mathbf{E}_Z + \mathbf{E}_Z \otimes \mathbf{E}_\Theta), \end{aligned} \quad (20)$$

$$\begin{aligned} \mathbf{b} = \overline{\mathbf{b}} = & (\lambda_\theta \lambda_z)^{-2} \mathbf{e}_r \otimes \mathbf{e}_r + (\lambda_\theta^2 + \gamma^2 \lambda_z^2) \mathbf{e}_\theta \otimes \mathbf{e}_\theta + \lambda_z^2 \mathbf{e}_z \otimes \mathbf{e}_z \\ & + \gamma \lambda_z^2 (\mathbf{e}_\theta \otimes \mathbf{e}_z + \mathbf{e}_z \otimes \mathbf{e}_\theta). \end{aligned} \quad (21)$$

These tensors show the geometric deformations that are needed to calculate stress in hyperelastic materials. From the reference configuration we use the strain tensors to understand the strain energy function.

4.3 Equilibrium Equations

After the mathematical equations showing how an artery deforms from its reference configuration, the authors explain how internal stresses balance the external loads such as stretching or pressure. This is shown by equilibrium equations which make sure that the artery stays stable under these conditions. In equation 22 we are introduced to the equilibrium equation for a solid at rest under no forces.

$$\text{div } \boldsymbol{\sigma} = \mathbf{0}, \quad (22)$$

This equation makes sure that there aren't any internal forces in play on the artery. In equation 23 we see how in cylindrical coordinate system, it simplifies to a single equation in the r direction.

$$\frac{d\sigma_{rr}}{dr} + \frac{(\sigma_{rr} - \sigma_{\theta\theta})}{r} = 0 \quad (23)$$

Equation 23 shows the importance of balancing stress, and it sets up for calculating pressure and wall stresses in the system. To solve radial stresses, equation 24 takes the previous equation and fixes it from the outwards surface to the internal.

$$\sigma_{rr}(\xi) = \int_{\xi}^{r_o} (\sigma_{rr} - \sigma_{\theta\theta}) \frac{dr}{r}, \quad r_i \leq \xi \leq r_o. \quad (24)$$

Assuming that the outer surface is stress free, the equation for the internal pressure is simplified using the isochoric components of stress, which gives us the required pressure to have the artery in its deformed shape. The authors also calculate the torsional moment in equation 27(2), and the axial force (28), by integrating stress components over the arterial wall.

$$p_i = \int_{r_i}^{r_o} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r}, \quad M_t = 2\pi \int_{r_i}^{r_o} \bar{\sigma}_{\theta z} r^2 dr, \quad (27)$$

$$N = 2\pi \int_{r_i}^{r_o} \left[\int_{r_o}^{\xi} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r} - \bar{\sigma}_{rr} + \bar{\sigma}_{zz} \right] \xi d\xi. \quad (28)$$

Finally, the axial force which is then adjusted for internal pressure is shown in equation 29. This is useful for their inflation testing as it shows accurately the extra axial force being implemented beyond internal pressure.

$$F = \pi \int_{r_i}^{r_o} (2\bar{\sigma}_{zz} - \bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) r dr \quad (29)$$

5.0 Constitutive Model for Arterial Walls

The model proposed by the authors has arterial layers, media and adventitia as a fiber composite, showing us the anisotropic behavior of the collagen fibers. In equation 57 we are introduced to the strain energy function, being split into isochoric and anisotropic parts to account for collagen fibers to not be active at low pressure and the non-collagenous matrix.

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{a}_{01}, \mathbf{a}_{02}) = \bar{\Psi}_{\text{iso}}(\bar{\mathbf{C}}) + \bar{\Psi}_{\text{aniso}}(\bar{\mathbf{C}}, \mathbf{a}_{01}, \mathbf{a}_{02}), \quad (57)$$

$\bar{\mathbf{C}}$ is the modified Cauchy Green Tensor and $\mathbf{a}_{01}$ and $\mathbf{a}_{02}$ are the vectors that determine the fiber's direction. The anisotropic part of the equation shows the stiffening of the collagen during high strains, while the isotropic part shows the matrix material. The model also uses invariants of $\bar{\mathbf{C}}$ and structure tensors in equations 58 and 59 to find the fibers stretch.

$$\bar{I}_1(\bar{\mathbf{C}}) = \text{tr } \bar{\mathbf{C}}, \quad \bar{I}_2(\bar{\mathbf{C}}) = \frac{1}{2}[(\text{tr } \bar{\mathbf{C}})^2 - \text{tr } \bar{\mathbf{C}}^2], \quad \bar{I}_3(\bar{\mathbf{C}}) = \det \bar{\mathbf{C}} = 1, \quad (58)$$

$$\bar{I}_4(\bar{\mathbf{C}}, \mathbf{a}_{01}) = \bar{\mathbf{C}} : \mathbf{A}_1, \quad \bar{I}_5(\bar{\mathbf{C}}, \mathbf{a}_{01}) = \bar{\mathbf{C}}^2 : \mathbf{A}_1, \quad (59)$$

In equations 58 there are the standard invariants, while in equations 59 we have the invariants that find the fiber's stretch along their direction while under tension. In equations 59, $\bar{I}_4$ and $\bar{I}_5$ are the square of the stretches in directions $\mathbf{a}_{01}$, $\mathbf{a}_{02}$, and $\mathbf{A}$ is the tensors structure which is found by performing the dyadic product of the direction vectors $\mathbf{a}_{01}$, $\mathbf{a}_{02}$. From this we can reduce the model in equations 59 to simplify the strain energy function to depend on the important invariants, which is shown in equation 62.

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) = \bar{\Psi}_{\text{iso}}(\bar{I}_1, \bar{I}_2) + \bar{\Psi}_{\text{aniso}}(\bar{I}_1, \bar{I}_2, \bar{I}_4, \dots, \bar{I}_8). \quad (62)$$

This equation shows the importance of critical matrices such as I1 and I2 for the isotropic element, and I4 and I6 for the stretching of collagen fibers. This model further simplifies into equation 63:

$$\bar{\Psi}(\bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) = \bar{\Psi}_{\text{iso}}(\bar{I}_1) + \bar{\Psi}_{\text{aniso}}(\bar{I}_4, \bar{I}_6). \quad (63)$$

I2 was dropped because it wasn't needed as the invariants used are enough to simulate the typical features of the arterial tissue. The isotropic neo-Hookean model is shown in equation 64:

$$\bar{\Psi}_{\text{iso}}(\bar{I}_1) = \frac{c}{2}(\bar{I}_1 - 3), \quad (64)$$

This models the elastin deformation in the matrix, where variable c is the specific layer's stiffness. The I1-3 in parenthesis is the part of the equation that makes sure that zero energy in the undeformed state. The effect of each layer creates the need to have exponential form for the strain energy in the fibers, as shown in equation 65.

$$\bar{\Psi}_{\text{aniso}}(\bar{I}_4, \bar{I}_6) = \frac{k_1}{2k_2} \sum_{i=4,6} \{ \exp[k_2(\bar{I}_i - 1)^2] - 1 \}, \quad (65)$$

In this equation the I1 part makes sure that the fibers are active in only tension. The exponential then catches the stiffening when in tension. The k variables represent the media and adventitia fibers that don't affect the arteries' mechanical response.

$$\bar{\boldsymbol{\sigma}} = c \text{dev} \bar{\mathbf{b}} + \sum_{i=4,6} 2\bar{\Psi}_i \text{dev}(\mathbf{a}_i \otimes \mathbf{a}_i), \quad (66)$$

The strain energy model is then transitioned to equation 66 which shows us the stresses explained by the isochoric Cauchy stress $\boldsymbol{\sigma}$. B is the left Cauchy green tensor with cdevb representing the isochoric component. The second term in the equation represents the anisotropic

collagen components, where $\mathbf{a}$ is the fiber directions, and the deviatoric operator represents the sensitivity of fiber stress.

The article states that to show the constitutive model equations, they had to study a healthy arterial segment. The model shows two separate strain energy functions for the media and adventitia layers, that were then modeled as a thick-walled 2 layered tube. The equations 67 and 68 shown below are then expanded to show their framework.

$$\begin{aligned}\bar{\Psi}_M &= \frac{c_M}{2}(\bar{I}_1 - 3) + \frac{k_{1M}}{2k_{2M}} \sum_{i=4,6} \{\exp[k_{2M}(\bar{I}_{iM} - 1)^2] - 1\}, \\ R_i &\leq R \leq R_i + H_M,\end{aligned}\tag{67}$$

$$\begin{aligned}\bar{\Psi}_A &= \frac{c_A}{2}(\bar{I}_1 - 3) + \frac{k_{1A}}{2k_{2A}} \sum_{i=4,6} \{\exp[k_{2A}(\bar{I}_{iA} - 1)^2] - 1\}, \\ R_i + H_M &\leq R \leq R_o\end{aligned}\tag{68}$$

Expanded Media and Adventitia, Respecitvely:

$$\bar{\Psi}_M = \frac{c_M}{2}(\bar{I}_1 - 3) + \frac{k_{1M}}{2k_{2M}} [\exp(k_{2M}(\bar{I}_{4M} - 1)^2) + \exp(k_{2M}(\bar{I}_{6M} - 1)^2) - 2]$$

$$\bar{\Psi}_A = \frac{c_A}{2}(\bar{I}_1 - 3) + \frac{k_{1A}}{2k_{2A}} [\exp(k_{2A}(\bar{I}_{4A} - 1)^2) + \exp(k_{2A}(\bar{I}_{6A} - 1)^2) - 2]$$

The models have a neo-Hookean term that shows the elastin matrix, where C_m represents the non-collagenous matrix, and C_a is the isotropic part of the tissue response. The exponential terms show the fiber being activated when it is stretched beyond its resting length, and the summation accounting for the 2 symmetric collagen fiber families. The fiber alignment is shown in equation 69 & 70 where we see the direction vectors for the collagen fibers being used once again.

$$\mathbf{A}_{1j} = \mathbf{a}_{01j} \otimes \mathbf{a}_{01j}, \quad \mathbf{A}_{2j} = \mathbf{a}_{02j} \otimes \mathbf{a}_{02j}, \quad j = M, A, \tag{69}$$

$$[\mathbf{a}_{01j}] = \begin{bmatrix} 0 \\ \cos \beta_j \\ \sin \beta_j \end{bmatrix}, \quad [\mathbf{a}_{02j}] = \begin{bmatrix} 0 \\ \cos \beta_j \\ -\sin \beta_j \end{bmatrix}, \quad j = M, A, \quad (70)$$

6.0 Results & Discussion

Presented below are the COMSOL simulation graphs and some of the corresponding graphs in the article.

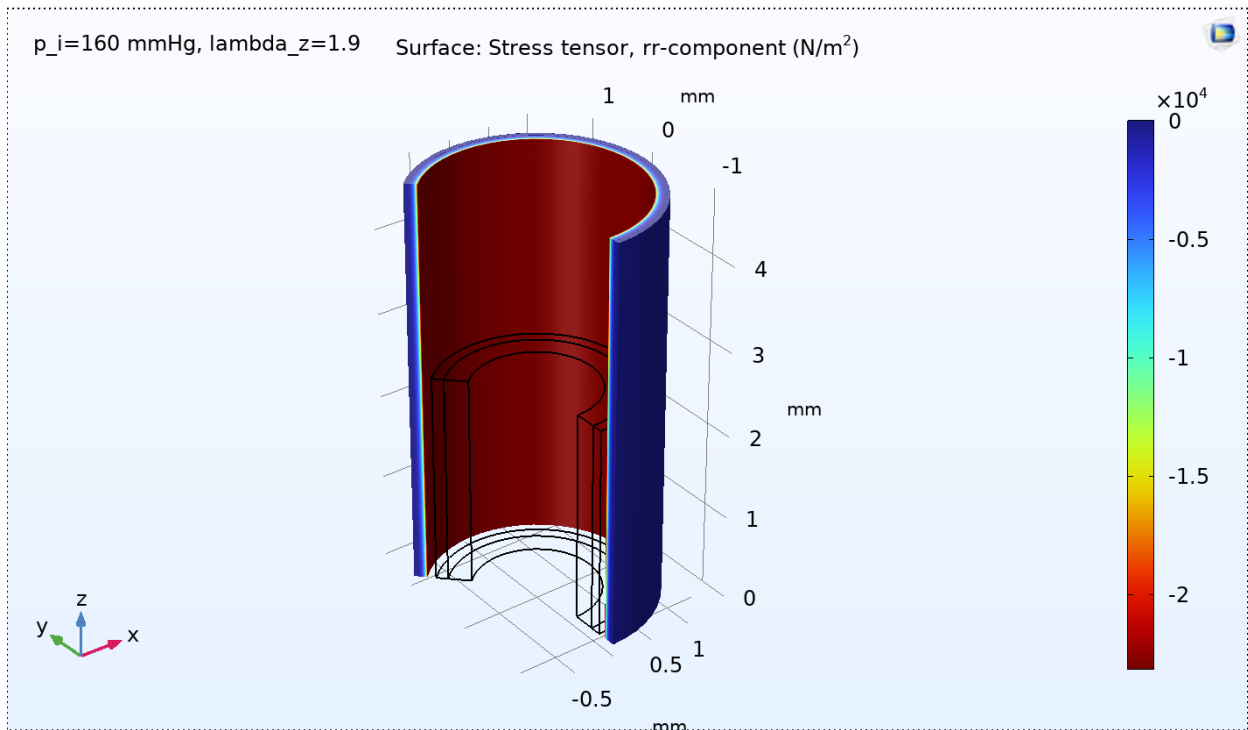


Figure 4: Radial Stress Distribution in artery wall at an axial stretch of 1.9 and 160 mmHg of internal pressure

In Figure 4, I simulated the stress distribution for an arterial wall that has an axial stretch of $\lambda = 1.9$ and an internal pressure of 160 mmHg. Radial stress decreases from the inner to outer radius as shown above. This is something that was expected because when the internal surface

experiences compressive stress, the outer layer remains at equilibrium. This distribution further shows how the arterial walls layers are anisotropic.

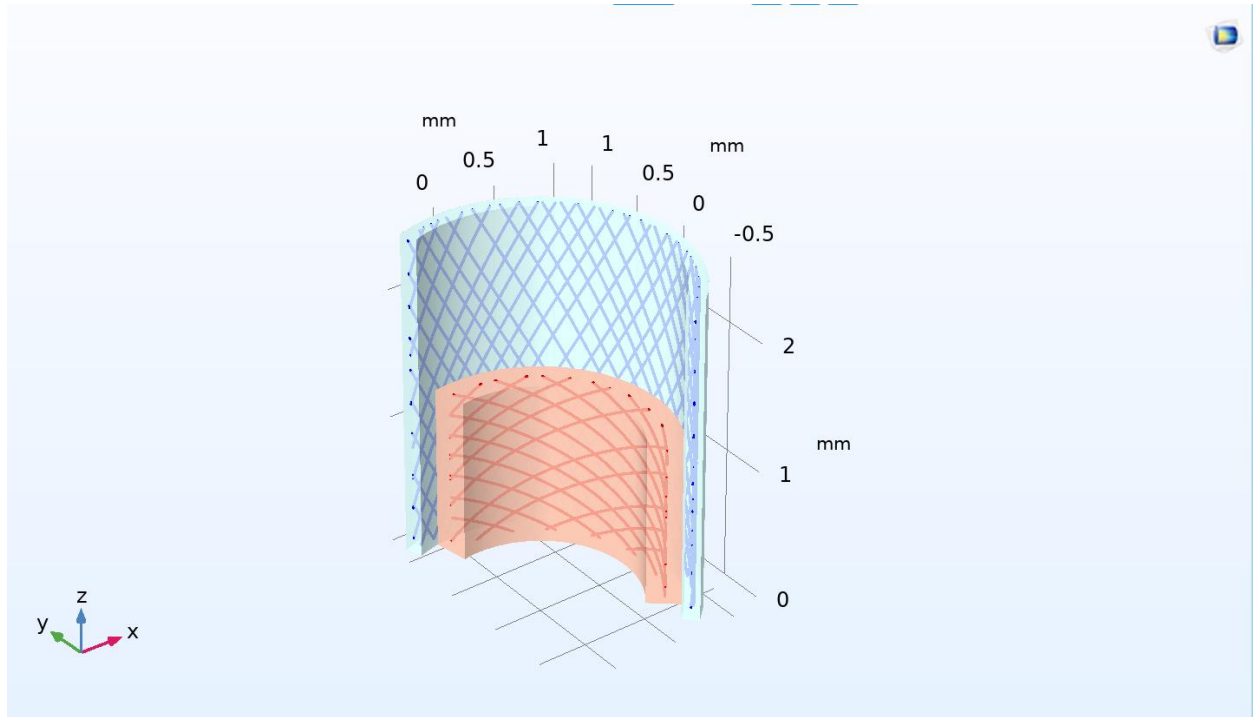


Figure 5: Fiber layout in the undeformed configuration of the media (inner, red) and the adventitia (outer, blue)

In Figure 5, I simulated the collagen fiber layout in the undeformed configuration for the 2 important layers, the media and adventitia. The media layer shown in the red and adventitia is shown in the blue, each have different fibers at different angles which was shown in the code provided for the simulation. The model does well to help capture the collagen structure.

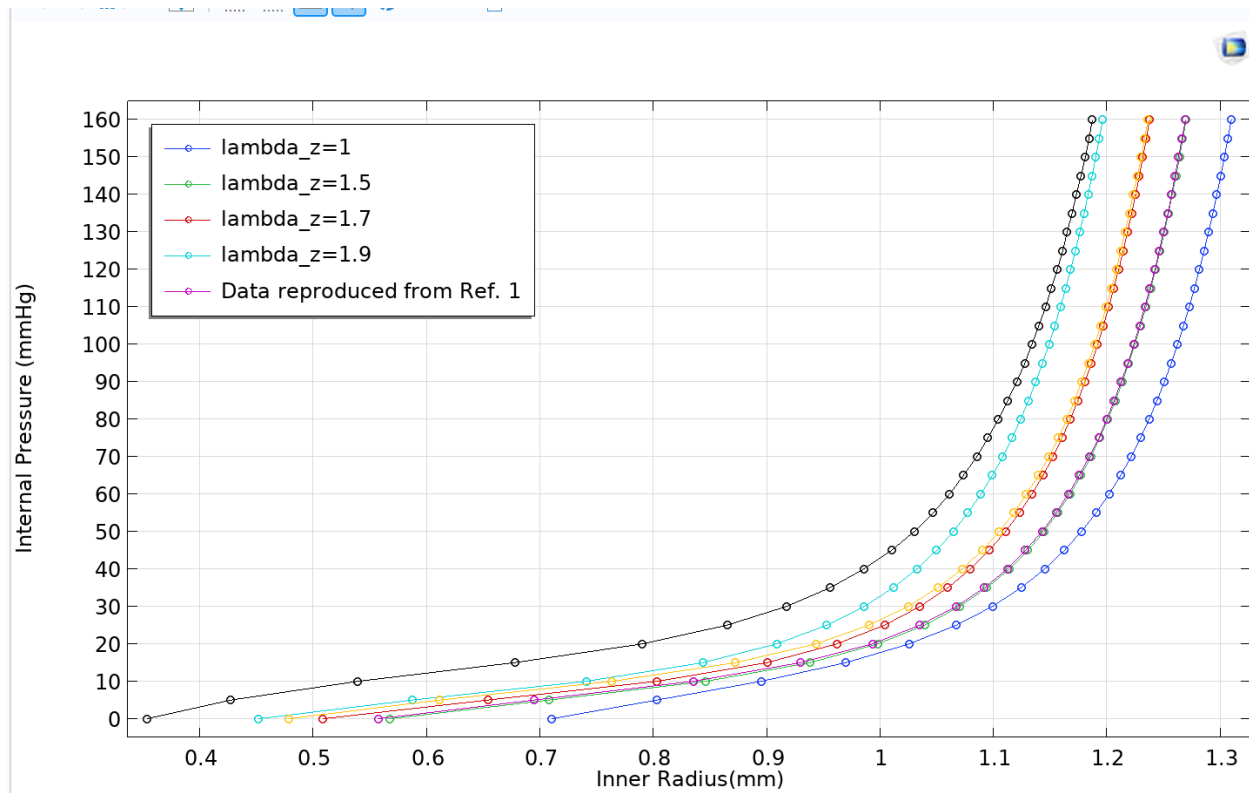


Figure 6: Plot of internal pressure vs. inner radius for 3 different axial stretches.

Figure 6 is important because it is our internal pressure vs inner radius graph. This plot is for 4 different axial stretches and it shows that as axial stretches increase, the artery does become stiffer as more pressure is introduced for a smaller increase in radius. This simulations result are aligned with the article's experimental data as seen in Figure 16a below. This shows the model being a good way to simulate pressure vs. inner radius under loads.

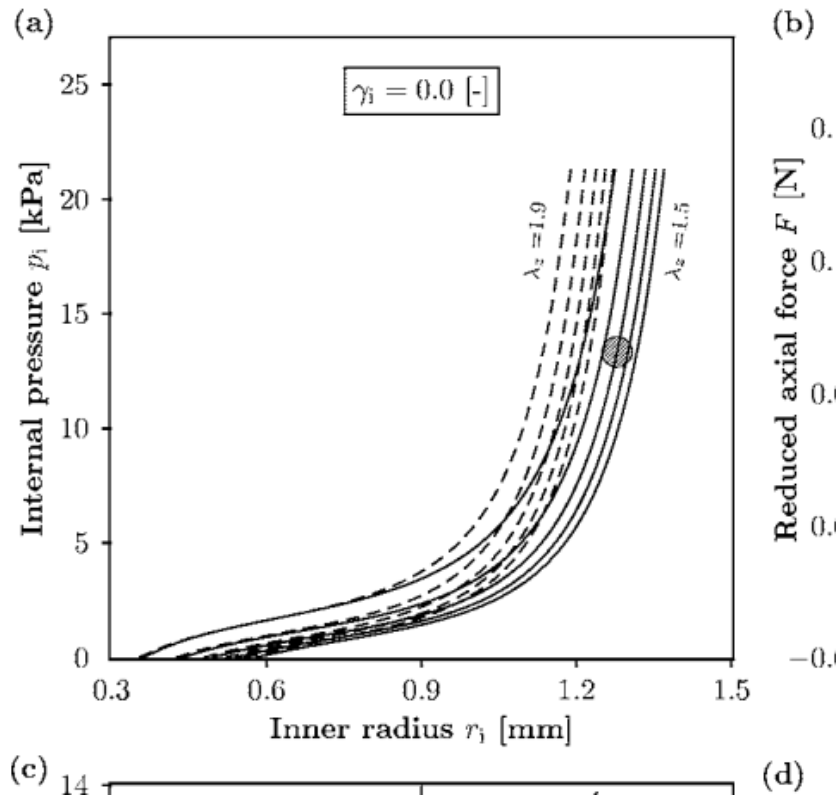


Figure 16. Deformation behavior of a carotid artery during inflation and torsion using the constitutive model (67)–(68). Solid lines are numerical results with residual strains included ($\alpha = 160.0^\circ$) and the dashed lines are results without residual strains ($\alpha = 0.0^\circ$). Dependence of (a) the internal pressure p_i and (b) the reduced axial force F on inner radius r_i , without shear deformation ($\gamma_i = 0$). Dependence of (c) the torsional couple M_i and (d) the reduced axial force F on the shear γ_i at fixed internal pressure $p_i = 13.33$ [kPa]. The shaded circles indicate the approximate central region of the physiological state.

As you can see from the articles graph in figure 16a, we see a bunch of plots showing the artery during inflation due to internal pressure. The solid curves use residual stress, and dashed curves show response with no residual stress. The internal pressure vs. inner radius curve is almost identical to the simulated one, proving that the model shows the arterial behavior.

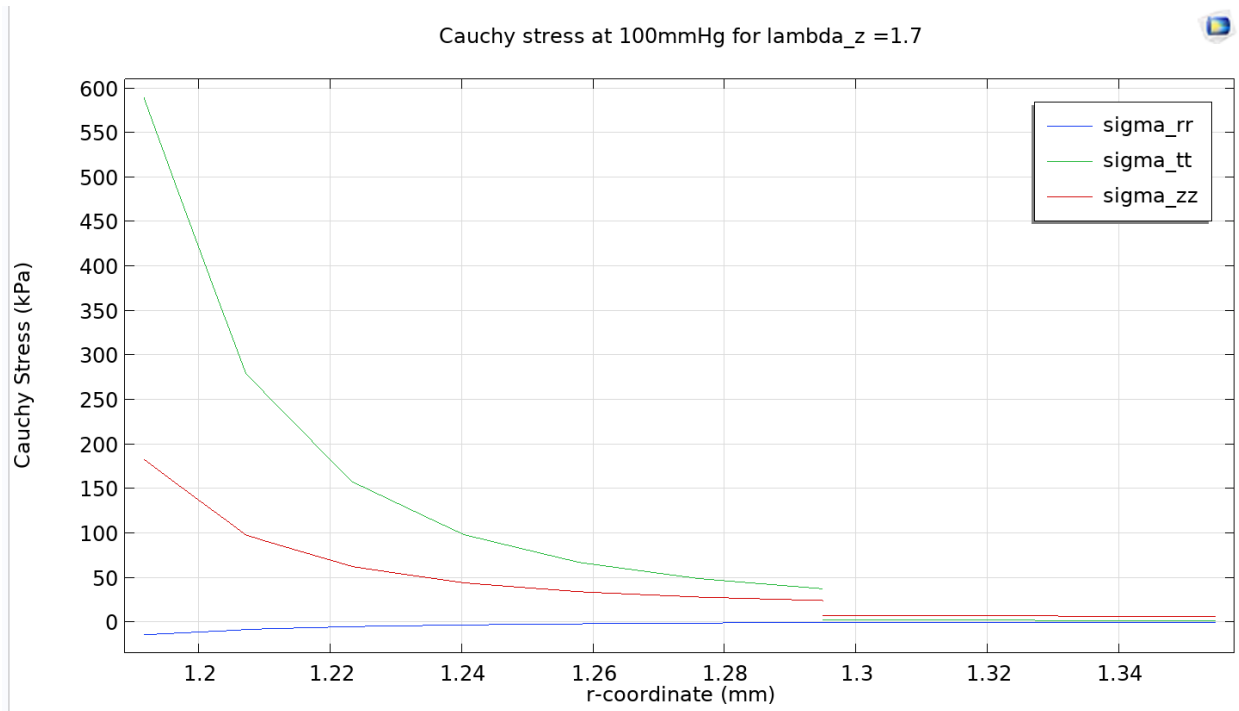


Figure 7: Cauchy stress at 100mmHg for $\lambda_z = 1.7$

In Figure 7, I simulated the Cauchy stress distribution for 100mmHg at $\lambda = 1.7$. It shows how the Cauchy stress is higher closer to the inner surface than further away from it. σ_{tt} and σ_{zz} stresses decrease to 0 while σ_{rr} remains mostly constant. This is important because it shows that the strain energy in this area is higher because of the internal pressure in the artery.



Figure 17. Mechanical separation of the layers of a human external iliac artery into a stiff media-intima tube (on the left-hand side) and a limp adventitia (from C. Schulze-Bauer, MD, Computational Biomechanics, Graz University of Technology, Austria, with permission).

Figure 17 shows the mechanical separation of arterial wall layers. This image shows how important it is to model each layer with its own behavioral properties. The media looks like a tube (shown in the simulation) and adventitia is on the right.

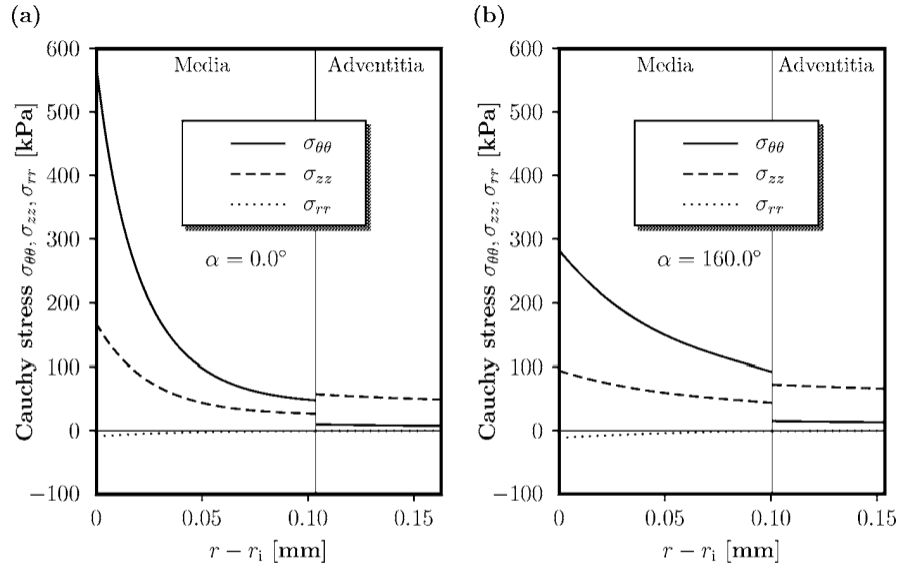


Figure 18. Plots of the principal Cauchy stresses $\sigma_{\theta\theta}$, σ_{zz} , σ_{rr} in the circumferential, axial and radial directions through the deformed media and adventitia layers in the physiological state with $p_i = 13.33$ [kPa], $\lambda_z = 1.7$, $\gamma_i = 0$: (a) without residual stress ($\alpha = 0.0^\circ$); (b) with residual stress ($\alpha = 160.0^\circ$). The abscissa is $r - r_i$. The numerical results are obtained for the constitutive models (67) and (68) with geometrical data and material constants as in Figure 14.

Figure 18 shows the principal stress components in various directions through the media and adventitia layers. In figure 18a we see the plot without residual stress and in 18b a plot with residual stress. The difference between the two is how in 18b the sigma tt and zz are reduced at the inner wall and in the media the slope is much steeper for the graph without residual stress. This is compared to the simulated graph in figure 7 where we see the 2 stresses decreasing and as they hit the inner wall on the simulation, they go to 0, while in the model they are separated. This could be due to a simulation error where sigma zz doesn't increase after it reaches the inner wall between the 2 layers. The other stresses are identical to the articles' plot.

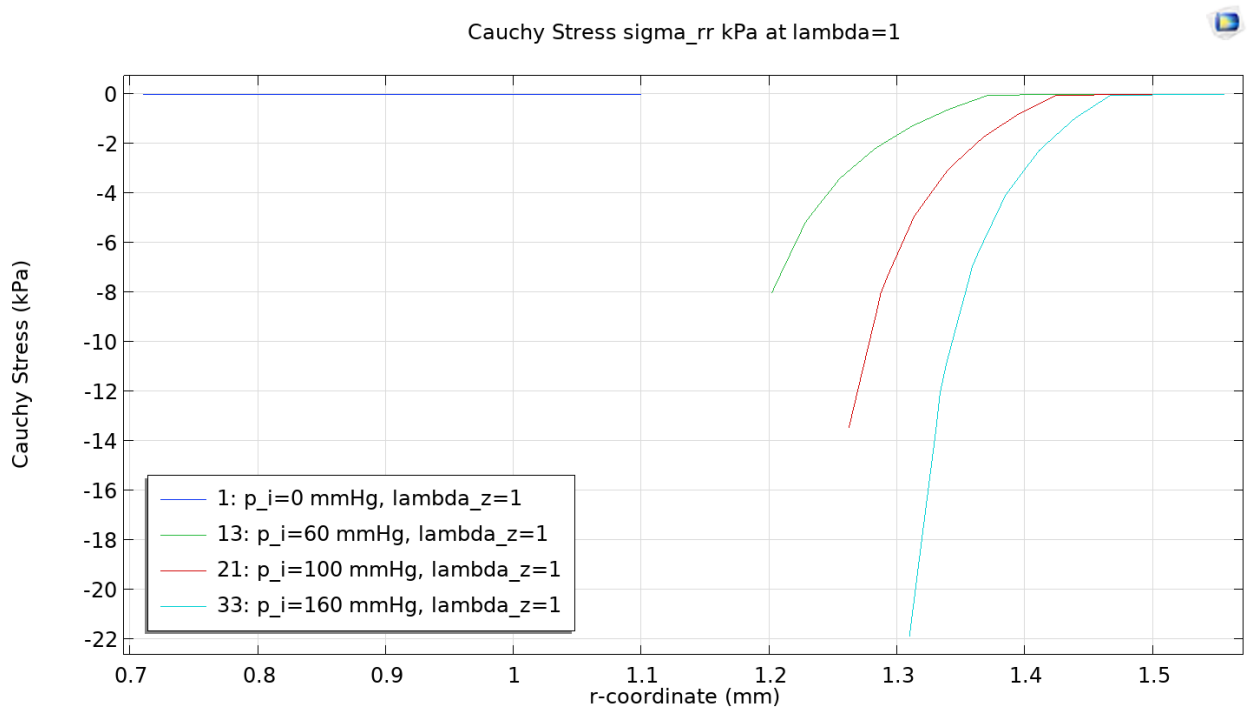


Figure 8: Cauchy stress for σ_{rr} at 60,100,160 mmHg for $\lambda_z=1$

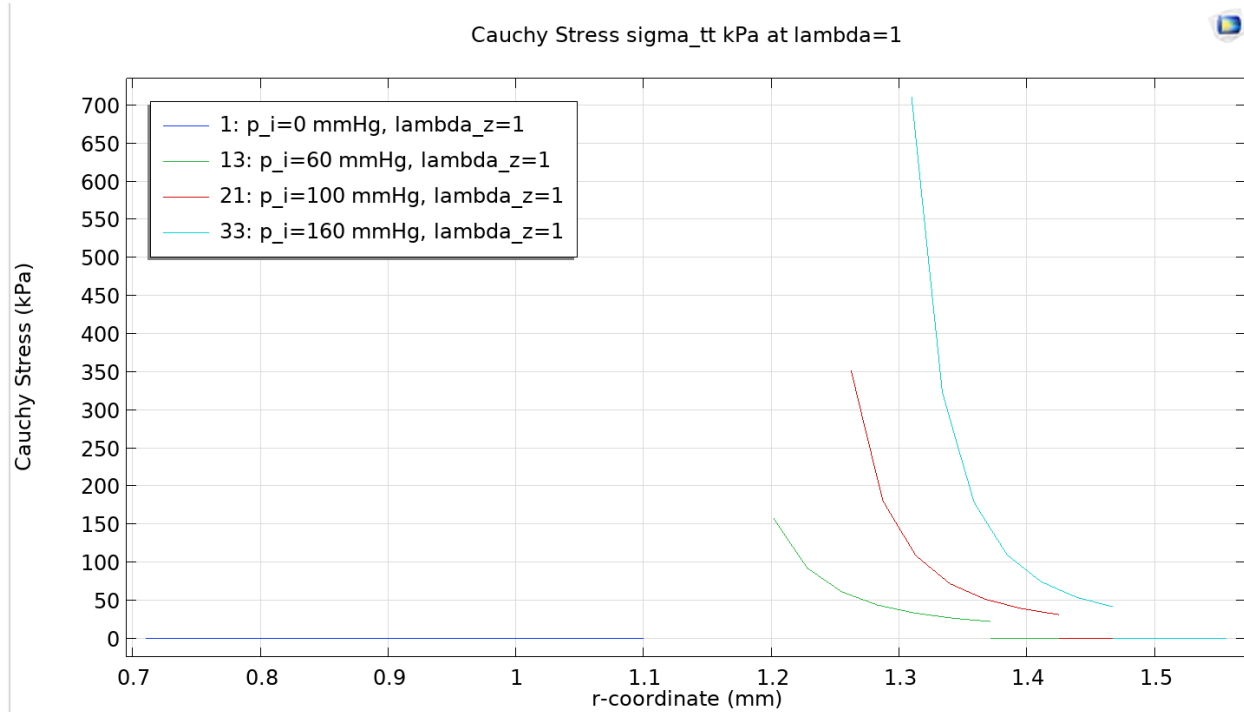


Figure 9: Cauchy stress for σ_{tt} at 60,100,160 mmHg for $\lambda_z=1$

Finally, in figures 8 and 9 we have the radial stress and circumferential stress components for 3 different pressure quantities. The λ remains the same at 1. Something that I notice in these plots is that they show symmetry in the data. While σ_{rr} has an exponential increase with the radius, while σ_{tt} has an exponential decrease with an increasing radius. The curves appear to be inverses of each other. This shows how for rr it starts with the negative stress values at the inner wall and then goes to 0, while the tt stress is max at the inner wall and then goes to 0 at the end. This difference shows how the arterial wall separates pressure across different stress modes.